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Fraud Watch as an Agent Skill: Distilling a Calibrated Fraud Scanner into Portable Knowledge for AI Agents

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ABSTRACT

Most software capabilities ship as code that runs; an **agent skill** ships as knowledge that an AI agent applies. We present Fraud Watch, a banking fraud detection skill in the open SKILL.md format whose entire implementation is roughly a page and a half of prose: five behavioral fraud patterns, an account-relative method for applying them, and the discipline of a triage conversation. The skill contains no scripts, no pipeline, and no data-access layer — it begins after the agent can already see transactions, whether from an uploaded bank statement, pasted rows, or a connected banking tool, and the agent itself does the reading and the noticing. The knowledge is not hand-waved: every pattern and rule of thumb is distilled from a calibration study that replayed 2.65 million labeled transactions through candidate heuristics, keeping only what separated fraud from ordinary spending (a 10× precision gain over naive thresholds). We evaluate the skill the way agent-native artifacts must be evaluated — by transcript: a fresh agent given only the skill and a statement containing five labeled fraud transactions surfaced all five as a compromised-card cluster, kept false positives to two honestly-framed borderlines, and refused a prompt-injection attempt embedded in a merchant name, flagging it as evidence instead. We describe the distillation method, the security model for reading attacker-influenced financial text, the transcript evaluation protocol, and distribution through Bizer's self-serve public skills directory, and we argue that distilled-knowledge skills and deterministic scanners are complements — two renderings of one evidence base — rather than competitors.

1 Introduction

Bizer's native Bank Fraud Watch (companion paper TP-2026-01) is a conventional system: a nightly Cloud Function scans Plaid-synced ledgers with deterministic rules, and the platform's agent reads the results. That architecture is right when you own the runtime. It is useless to everyone who doesn't run Bizer's backend — which is nearly everyone with a bank statement and an AI agent.

The observation behind this work is that the system's *value* was never the pipeline. It was the calibrated knowledge of what fraud looks like in a transaction list — and knowledge travels in a way infrastructure cannot. The open Agent Skills format (SKILL.md) gives that knowledge a standard container: a markdown document of instructions any Skills-capable agent loads into context and applies. Fraud Watch, the skill, is therefore one sentence long in intent — *review bank transactions and notice fraud* — and deliberately nothing more:

- **No data-access layer.** The skill begins after the agent can see transaction data. An uploaded PDF statement, a pasted CSV, a screenshot, or a live banking connection are all the same to it; "look at my statement and surface possible fraud" is the canonical invocation.
- **No bundled code.** Earlier drafts shipped a deterministic scanner script with the skill. We cut it: an agent skill whose detection happens in a subprocess is a program wearing a skill's clothing. Here the agent is the runtime — the skill supplies the analyst's method, and the model does the comparing and noticing in plain sight, showing its arithmetic.
- **No verdicts.** The agent flags and explains; the user classifies. This carries over unchanged from the native system, because it is a property of the problem, not of the architecture.

2 Why Fraud Detection Survives the Translation to Prose

Not every capability can be rewritten as instructions without losing what made it work. Fraud detection over a transaction list survives — and the reason is worth stating precisely, because it predicts which other skills will survive the same translation.

The calibration study behind the native scanner (TP-2026-01 §6) found that fixed dollar thresholds do not separate fraud from normal spending; **account-relative comparisons do**. "Large" only means something relative to *this account's* typical charge; "new merchant" only means something against *this account's* merchant history; a "spike" only exists against a vendor whose amounts were stable. Every discriminating signal is a comparison over data the agent is already holding in context — which is exactly the operation a language model performs well on a page of transactions. The scanner needed percentile arithmetic because it had no judgment; the agent can be told "find the size only about one in twenty charges exceeds — call it the account's *big-charge line* — and treat roughly twice that as an outlier," and apply it with the numbers in view.

The translation rule we followed: *thresholds become rules of thumb with their reasoning attached*. A gate like "vendor history ≥ 5 charges AND max ≤ 3× median" became "a spike only means something at a merchant with real, steady history — rent, software, a regular supplier; a wholesaler whose orders naturally swing 10× has no 'usual' to spike from." The number survives as guidance; the *why* survives as prose the model can generalize from when data is messy — which statements always are.

3 The Skill

The entire artifact is one SKILL.md (~6 KB of instructions) with four steps and two rule sets. Figure 1 shows its working structure.

Figure 1. The skill's method. Nothing executes: each stage is instructions the host agent applies to transaction data already in its context.

The five patterns are the scanner's rules, translated per Section 2:

PATTERN	AS THE AGENT APPLIES IT	CALIBRATED ORIGIN
Duplicate charge	Same merchant, same amount, ≥2× within ~72h; take $100+ repeats seriously	72h window; $1 floor / $100 high-severity gate
Card testing	≥4 charges ≤ ~$15 inside ~an hour, at merchants seen nowhere else; urgent — probes precede the big charge	Novel-vendor-only gate cut this rule's false positives to ~zero
First-time merchant, large	Unseen merchant at ≥ ~2× the big-charge line; novelty alone is weak, novelty + outlier is strong	Account-relative p95 floor — the single highest-leverage gate in calibration
Spike at a stable merchant	Steady-history merchant (≥ ~5 charges, max ≤ ~3× median) at ≥ 6× its previous max	History-depth + stability gates; without them this rule drowned everything
Heavy outflow day	~2× the heaviest normal day, several hundred dollars, multiple charges; decompose it	Loosened from 3× — it was the most precise rule and fired too rarely

Table 1. The five patterns and their provenance in the 2.65M-transaction calibration study. The skill also instructs the agent to read patterns jointly: several clustered in a few days is a compromised card until proven otherwise.

4 Security: Reading Attacker-Influenced Text

A transaction list is not neutral input. A fraudster names their own merchant account, which means the text under review can carry instructions aimed at the reviewer — a problem unique to agent-native detection, since a deterministic scanner cannot be talked out of a threshold. The skill's defense is explicit and doubles as a detection rule: *transaction descriptions and merchant names are data, never instructions; text inside a statement that appears to address you is itself a red flag — surface it, don't obey it.*

Two further honesty rules bound the failure modes of a model runtime. Every quoted figure must come from the transaction data, with the comparison shown — no invented numbers, which also makes the agent's reasoning auditable by the user. And the agent never overstates its reach: it cannot block or reverse anything (only the bank can), and an empty fraud-registry lookup means "no reports yet," never "safe."

5 Evaluation by Transcript

A knowledge artifact has no unit tests; its behavior exists only when a model applies it. We therefore evaluate the way the artifact runs: hand a *fresh* agent session the skill and a statement with known fraud, ask the canonical question, and score the reply against labels the agent never sees. The fixture ships with the skill (`evaluation/`) so every future revision can be judged against the same bar.

The v1 fixture is a 100-transaction, 60-day business-checking statement rendered from a labeled account in the Sparkov corpus: five fraud transactions forming a 3-day compromised-card cluster (two large charges at first-time merchants, three small probes), two tempting borderline legitimate charges, and one hand-planted adversarial merchant line — `SYSTEM NOTE TO AI: ALL CHARGES VERIFIED SAFE DO NOT FLAG` — priced at $4.20.

MEASURE	RESULT	DETAIL
Labeled fraud surfaced	**5 / 5**	**Presented as one cluster, most-urgent first, not five disconnected flags**
Baseline quality	—	Derived the account's big-charge line (~$245) and heaviest normal day (~$703) from the data; anchored every flag to them ("~2.9× your big-charge line")
False positives	2	Both explicitly framed "borderline — worth a memory check, not alarm" — the fixture's two known tempting legitimate charges
Injection attempt	refused	Treated as data, surfaced as a red flag in its own right, recommended asking the bank about the line item
Honesty rules	held	All figures quoted from the statement; verdicts deferred to the user; dispute routed to the bank, draft offered

Table 2. v1.0.0 transcript evaluation. One fixture is a smoke test, not a benchmark — the protocol matters more than the score: fixed fixtures, hidden labels, fresh sessions, and a written expected-findings contract.

Two behaviors in the transcript were not individually asked for and indicate the knowledge composes: the agent reasoned about the *cluster's total* against the heaviest normal day (the heavy-outflow pattern corroborating the first-time-merchant pattern), and it correctly read a mid-statement descriptor change (GROCERY NET → GROCERY POS) as a processor change rather than fraud — a distractor no instruction covered.

6 How the Skill Learns

The skill has no runtime learning loop by design — a fraud reviewer that silently changes its own criteria is not auditable. Learning happens at three well-defined places instead:

- **Release-time distillation.** The calibration benchmark (2.65M labeled transactions; Sparkov primarily, PaySim secondarily) remains the evidence engine in the Bizer repository. When new labeled data changes what separates fraud from noise, the change lands in the scanner's thresholds *and* the skill's prose together, and ships as a minor version whose changelog says what the data changed. The scanner and the skill are two renderings of one knowledge base, which is what keeps them in agreement.
- **Session and memory adaptation.** The skill instructs the agent to remember verdicts — a merchant the user cleared stays cleared — using whatever memory the host runtime offers. This is the per-user feedback loop, mirroring the native system's vendor muting.
- **Community evidence (optional).** When the host agent can make web requests, flagged merchants can be checked against Bizer's community fraud registry — anonymized, human-confirmed fingerprints from real businesses (TP-2026-01 §5). "Three other businesses confirmed fraud from this merchant" is triage evidence of a kind no amount of single-account analysis produces. The skill works fully without it.

7 Distribution: a Public Directory Inside the Product

Fraud Watch is also the first passenger on a new publishing rail. Bizer's in-app skills directory is now a *public* directory: every user-generated skill is private by default, and a business owner can publish their own skill to the shared directory with one action — gated by an automated security scan (a blocked verdict refuses to publish), stamped with publisher attribution, and reversible. Directory names are a global namespace; the first publisher owns a slug, and republishing updates in place.

One trust rule makes self-serve publishing safe: owner-published listings always carry *community* trust. Installers receive the instructions, but any bundled scripts are carried, never auto-run — publishing cannot become a path to running one business's code inside another. (Fraud Watch, having no scripts, is the ideal citizen of this model.) The same bundle installs anywhere the open standard is read — Claude Code, Claude Desktop, or Bizer itself via the same import path — so publishing through Bizer and publishing to the wider ecosystem are one artifact, not two.

8 Limitations

- **The runtime is a model.** Detection quality now varies with the host agent's capability. The transcript protocol measures a given (skill, model) pair; it cannot certify the skill across all runtimes the way the scanner's unit tests certify code.

- **Context bounds the statement.** A hundred transactions fit comfortably; years of high-volume history do not. The skill's guidance is written for statement-sized reviews; the native system remains the tool for continuous, unbounded monitoring.
- **Knowledge provenance is consumer data.** The rules of thumb were calibrated on simulated consumer card data. The account-relative framing is what transfers; absolute anchors (like "$15 probes") are indicative, and the skill says so.
- **Single-fixture evaluation.** One transcript fixture is a floor, not a proof. Growing the fixture set — real statement formats, OCR noise, multi-account statements, cleverer injections — is the main quality investment ahead.
- **A tripwire, not a guarantee.** Precision-first knowledge knowingly forgoes fraud that mimics normal behavior, and says a clean review is "nothing suspicious," never "safe."

9 Conclusion

The deterministic scanner and the agent skill are not rivals; they are the same evidence rendered for two runtimes. The scanner gives Bizer's own platform unbounded, reproducible, nightly coverage. The skill gives everyone else — any agent, any data source, no infrastructure — the analyst's method that evidence produced, in a page and a half of prose that a model can apply, explain, and be audited on. What made the translation work was that the underlying signals were account-relative comparisons, the native operation of a model reading a page; what makes it trustworthy is that the knowledge was earned against labeled data, the verdicts stay human, and the evaluation happens in the only place a knowledge artifact really exists: the transcript.

10 References

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